1) $A \rightarrow B$ 1st order $10^{-4}\ mol/g\cdot s = r_A$

a) The rate of stirring is so high that the stirring effect is negligible.

b) $$C_{WP} = \frac{-r_A \rho_c R^2}{D_e C_{AS}}$$ $C_{WP} \lll 1 \Rightarrow$ No pore diffusion limitation

$\rho_c = 2\ g/cm^3$, $R = 0.01$, $D_e = 0.1\ cm^2/s$, $C_{AS} = 10^{-3}$

$10^{-4}\ mol/g\cdot s = -r_A$

c) PBR: $$F_{A0}\frac{dx}{dW} = -r_A'\eta = -k'C_{A0}(1-x)\eta$$

$$\eta = \frac{F_{A0}}{W k' C_{A0}}\left(-\ln(1-x)\right)$$

$$W = V \times \rho_B = \underbrace{\left(10\,m \times \frac{100\,cm}{m}\right) \times (5^2\,cm^2)\pi}_{V} \times \underbrace{\left(\frac{1\,g}{cm^3}\right)}_{\rho_B}$$

$W = 78539.8\ g$

$$k' = \frac{r_A}{C_{A0}} = 0.1\ \frac{L}{g\cdot s}$$

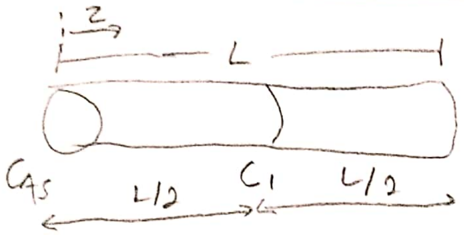
$$\eta = \frac{1}{(78539.7)(0.1)(1\times10^{-3})}\left(-\ln(1-.75)\right) = 0.1765$$

$$\Phi = \frac{1}{0.1765} = 5.67 \Rightarrow \phi = \frac{R}{3}\sqrt{\frac{k' \rho_c}{D_e}} \Rightarrow R = 0.38$$

$$C_{WP} = \eta \phi^2 = 0.1765 (5.67^2) = 5.67 >> 1$$

There is pure diffusion limitations

2)



a) $$W_A (\pi d^2/4)\big|_z - W_A (\pi d^2/4)\big|_{z+\Delta z} + r_A''(\cancel{\pi d}\Delta z) = 0$$

$$C(z) = Az + B$$

BC:

$$C(z=0) = C_{AS} = A(0) + B \Rightarrow \boxed{C_{AS} = B}$$

$$C(z = L/2) = C_1 = A\left(\frac{L}{2}\right) + B$$

$$C_1 = A\left(\frac{L}{2}\right) + C_{AS}$$

$$\boxed{\frac{C_1 - C_{AS}}{\left(\frac{L}{2}\right)} = A}$$

$$\boxed{C(z) = \frac{C_1 - C_{AS}}{\left(\frac{L}{2}\right)} z + C_{AS} \qquad 0 < z < L/2}$$

b) $\eta_1 = \dfrac{\tanh(\phi_1)}{\phi_1}$, $\phi_1 = L\sqrt{\dfrac{4k''}{dD_e}}$

$$\phi_{non\text{-}poisoned} = \left(\frac{L}{2}\right)\sqrt{\frac{4k''}{dD_e}} = \frac{\phi_1}{2}$$

$$\eta_{non\text{-}poisoned} = \frac{\tanh(\phi_1/2)}{\phi_1/2}$$

$$\eta \cdot (\cancel{\pi} d \cancel{L})(\cancel{k''} C_{as}) = \eta_{np}(\cancel{\pi} d \cancel{L}/2)(\cancel{k''} C_1)$$

$$\eta = \eta_{np} \cdot \frac{C_1}{2C_{as}} = \frac{C_1}{C_{As}} \frac{\tanh(\phi_1/2)}{\phi_1}$$

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